01 / PROBLEM AND CONTRIBUTION

Tiny solar-supply PCB adaptation

Independent hardware design · KiCad | PCB design study

DESIGN AT A GLANCE

Adapted a documented Elektor solar-supply circuit into a KiCad PCB using an AP3015 boost stage, Schottky diodes, and MOSFET-controlled shutdown.

Engineering question

How can a small solar-supply circuit be organized into a compact board with charging and shutdown paths?

Situation

The project README credits the Elektor Tiny Solar Supply article by Clemens Valens. The design targets a NiMH-cell solar supply with a nominal 3.3 V output, and the KiCad schematic uses an AP3015KTR-G1 controller.

Task

Adapt the referenced circuit into a schematic and PCB with battery, solar, control, and output connections.

My contribution

KiCad schematic and PCB adaptation of the Elektor Tiny Solar Supply circuit, credited in the project README to Clemens Valens.

Tools and concepts

KiCad / PCB layout / Boost-converter circuits / Solar power circuits / Schematic capture

02 / IMPLEMENTATION

Circuit to board

01 / Design step

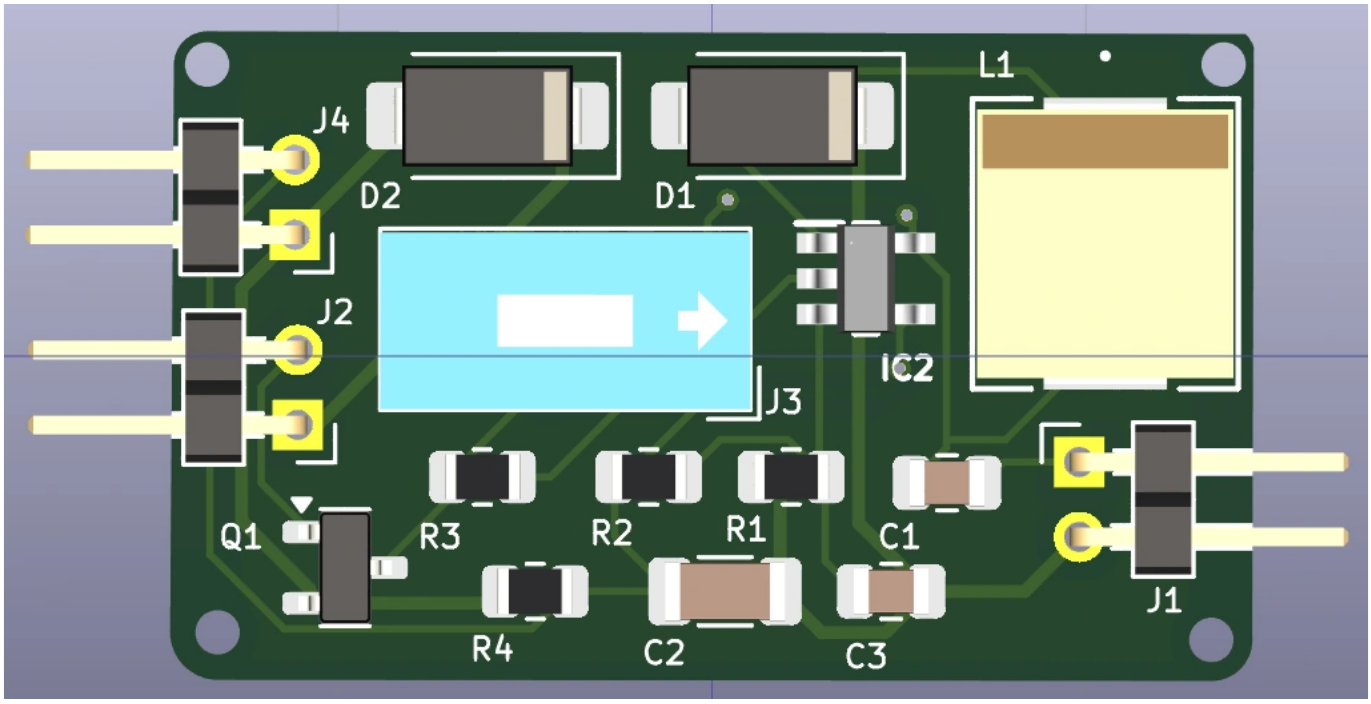
Implemented the boost-converter network, SS14 Schottky diodes, feedback components, and a 2N7000 shutdown-control device in the source schematic.

02 / Design step

Organized the external connections and component placement into a two-copper-layer board. The current PCB contains 16 footprints, 85 track segments, and four vias.

03 / Design step

Documented the schematic and board render while preserving the original circuit attribution. Described the nominal output and daylight shutdown as intended behavior, pending measurement.



Solar-supply board render

Repository 3D render of the adapted supply board.

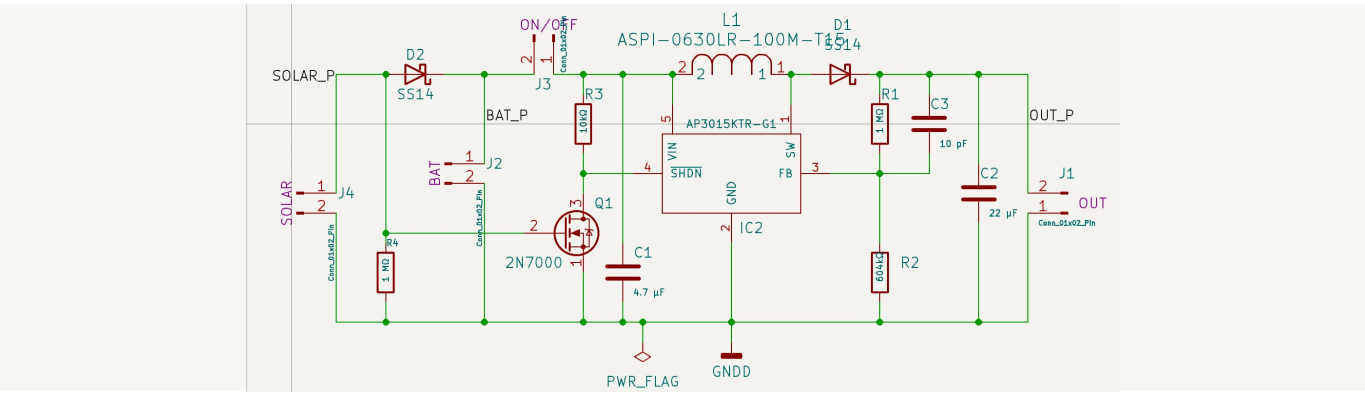
03 / EVIDENCE AND SCOPE

Design evidence and next steps

Current board	2 copper layers
Boost controller	AP3015KTR-G1
Nominal output intent	3.3 V

Available evidence

The repository contains an editable KiCad schematic and board, plus schematic and 3D previews. This supports a PCB adaptation case study; it does not establish output-current capability, charging performance, or conversion efficiency.



Solar-supply schematic

Repository schematic of the AP3015-based circuit; original concept credited to Elektor / Clemens Valens.

Scope and limitations

The circuit is an attributed adaptation. The collection overview's Li-ion/TP4056 description conflicts with this project's NiMH/AP3015 files and is not used. MPPT, measured efficiency, and a validated battery-charging profile are not claimed.

Next steps - proposed work

- Validate startup, shutdown, and output regulation under defined source/load conditions.
- Document battery-charging behavior and input/output power measurements.

Source: PCB-design-projects GitHub repository: Tiny-Solar-Power-Supply; README, KiCad schematic/board, and repository previews. Reviewed 11 November 2026.

Open the project repository